

PROBLEM STATEMENT

- A hidden fraudulent URL is an URL added to a trusted website that leads to an illegitimate web page
- **GOAL:** identify the characteristics that differentiate fraudulent URLs and URLs that are legitimate
- **LIMITATION OF THE STUDY:** imbalance between the two classes (hidden fraudulent and non-fraudulent URLs)

DATASET DESCRIPTION

Column Name	Description
url	The actual URL
compromissionType	Categorical value: "phishing," "defacement," or empty string.
isHiddenFraudulent	Boolean value: true if the URL is hidden and fraudulent.
contentLength	Integer value: Content-Length header obtained by an HTTP HEAD request to the URL.
serverType	String value: X-Server-Type header obtained as above.
poweredBy	String value: X-Powered-By header obtained as above.
contentType	String value: Content-Type header obtained as above.
lastModified	Date in the format "Thu, 17 Jan 2013 18:49:04 GMT": Last-Modified header obtained as above.

EXPLORATORY DATA ANALYTICS

- Initial exploratory analysis conducted in R to understand dataset characteristics and identify challenges
- Examined variable types using the `glimpse()` function from the "tidyverse" library
- Majority of variables were character type, except for `isHiddenFraudulent` (boolean) and `contentLength` (integer)

Variable	Distribution		
compromissionType	defacement	normal	phishing
	96514	3713	884953
isHiddenFraudulent	TRUE		FALSE
	8709		176471

- `compromissionType` exhibited similarities with `isHiddenFraudulent` in distribution
- Intra-class variance observed in variables like `lastModified`, `contentType`, `poweredBy`, and `serverType`.
- Large number of missing values discovered after converting empty cells to "NA"

NA per column			
<code>lastModified</code>	77592	<code>contentType</code>	2306
<code>serverType</code>	2082	<code>poweredBy</code>	83841

The figure displays six histograms arranged in a 3x2 grid, showing the distribution of various features. The y-axis for all plots is 'count'.

- CompressionType:** The x-axis has three categories: 'deflate', 'gzip', and 'none'. The 'deflate' category has the highest count, followed by 'gzip', and 'none' has a very low count.
- @ModerFraudRisk:** The x-axis has two categories: 'FALSE' and 'TRUE'. The 'FALSE' category has a very high count, while the 'TRUE' category has a very low count.
- contentType:** The x-axis shows a wide range of content types. The distribution is highly skewed, with the highest count for 'text/html' and a long tail of other content types.
- serverType:** The x-axis shows a wide range of server types. The distribution is highly skewed, with the highest count for 'Apache/2.4.18 (Ubuntu)' and a long tail of other server types.
- PowerBy:** The x-axis shows a wide range of power sources. The distribution is highly skewed, with the highest count for 'PowerBy' and a long tail of other power sources.

PREPROCESSING AND DATA MANIPULATION

- Addressed missing values:
 - removed observations with missing values for serverType and contentType attributes,
 - replaced missing values in poweredBy attribute with mean class value ("PHP")
- Removed lastModified attribute from the dataset
- Standardized and streamlined contentType and serverType attributes
 - Grouped server types by major names, ignoring version details
 - Divided contentType values into three major categories: text, picture, and application

PREPROCESSING AND DATA MANIPULATION

- Utilized pivot operator in RapidMiner to exclude observations with fractional counts < 0.01
 - Applied pivot operator on serverType and poweredBy attributes
 - Calculated fractional counts for each distinct value
 - Joined the resulting tables with the initial dataset to retain observations with substantial fractional counts
- Created new variables: URL length and protocol type

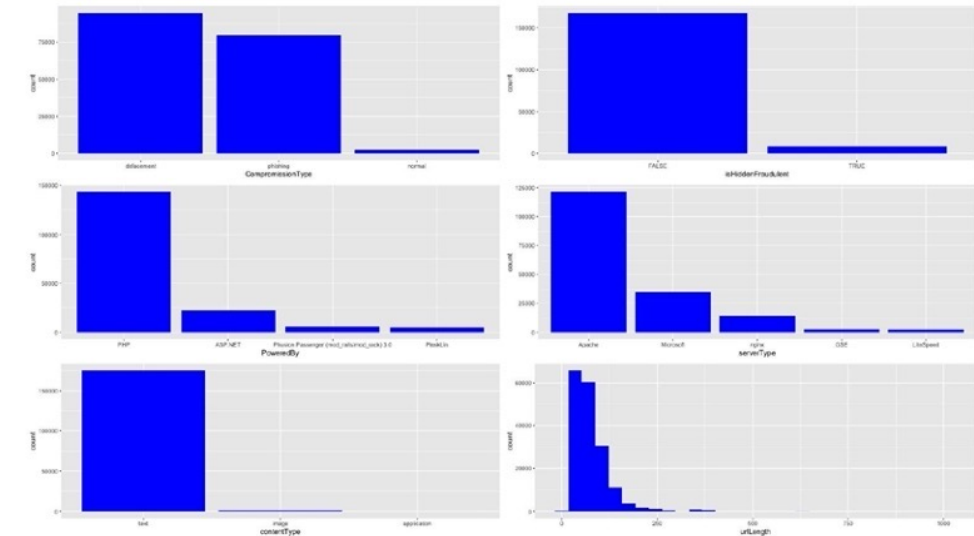
MODEL PREPARATION

- Created two parallel preprocesses for analyzing `isHiddenFraudulent` and `compromissionType`
- Defined target variable and positive class
- Converted nominal/categorical variables to numerical representations
- Reordered attributes and stored the preprocessed dataset
- Created a correlation matrix for a more comprehensive understanding of the data

CORRELATION MATRICES AND OTHER GRAPHS

Attributes	compr...	compr...	conten...	conten...	conten...	powere...	powere...	powere...	serverT...	serverT...	serverT...	serverT...	uIrLen...
compromissionType = defacement	1	-0.975	-0.144	0.080	-0.080	-0.363	0.374	-0.199	0.418	-0.144	-0.338	-0.138	0.235
compromissionType = phishing	-0.975	1	0.117	-0.078	0.078	0.362	-0.377	0.204	-0.411	0.148	0.333	0.141	-0.227
contentLength	-0.144	0.117	1	0.394	-0.394	0.217	-0.152	-0.044	-0.243	-0.032	0.340	-0.062	-0.137
contentType = image	0.080	-0.078	0.394	1	-0.993	-0.029	0.029	-0.016	0.041	-0.012	-0.038	-0.011	-0.057
contentType = text	-0.080	0.078	-0.394	-0.993	1	0.029	-0.029	0.016	-0.042	0.012	0.038	0.011	0.057
poweredBy = ASP.NET	-0.363	0.362	0.217	-0.029	0.029	1	-0.793	-0.070	-0.565	-0.051	0.760	-0.108	-0.196
poweredBy = PHP	0.374	-0.377	-0.152	0.029	-0.029	-0.793	1	-0.387	0.565	0.064	-0.571	-0.174	0.244
poweredBy = Phusion Passenger (...)	-0.199	0.204	-0.044	-0.016	0.016	-0.070	-0.387	1	-0.275	-0.025	-0.092	0.623	-0.161
serverType = Apache	0.418	-0.411	-0.243	0.041	-0.042	-0.565	0.565	-0.275	1	-0.200	-0.741	-0.442	0.167
serverType = GSE	-0.144	0.148	-0.032	-0.012	0.012	-0.051	0.064	-0.025	-0.200	1	-0.067	-0.040	0.380
serverType = Microsoft	-0.338	0.333	0.340	-0.038	0.038	0.760	-0.571	-0.092	-0.741	-0.067	1	-0.148	-0.228
serverType = nginx	-0.138	0.141	-0.062	-0.011	0.011	-0.108	-0.174	0.623	-0.442	-0.040	-0.148	1	-0.128
uIrLenght	0.235	-0.227	-0.137	-0.057	0.057	-0.196	0.244	-0.161	0.167	0.380	-0.228	-0.128	1

Attributes	conten...	conten...	conten...	isHidd...	powere...	powere...	serverT...	serverT...	serverT...	serverT...	uIrLen...
contentLength	1	0.457	-0.458	-0.193	0.186	-0.148	-0.184	-0.008	0.249	-0.029	-0.123
contentType = image	0.457	1	-0.993	-0.383	-0.024	0.018	0.030	0.004	-0.032	-0.003	-0.076
contentType = text	-0.458	-0.993	1	0.382	0.024	-0.018	-0.030	-0.004	0.032	0.004	0.075
isHiddenFraudulent = FALSE	-0.193	-0.383	0.382	1	-0.062	0.039	0.144	-0.001	-0.044	0.009	-0.201
poweredBy = ASP.NET	0.186	-0.024	0.024	-0.062	1	-0.830	-0.567	-0.036	0.747	-0.062	-0.165
poweredBy = PHP	-0.148	0.018	-0.018	0.039	-0.830	1	0.440	0.023	-0.602	0.078	0.157
serverType = Apache	-0.184	0.030	-0.030	0.144	-0.567	0.440	1	-0.231	-0.753	-0.458	0.034
serverType = LiteSpeed	-0.008	0.004	-0.004	-0.001	-0.036	0.023	-0.231	1	-0.048	-0.029	-0.025
serverType = Microsoft	0.249	-0.032	0.032	-0.044	0.747	-0.602	-0.753	-0.048	1	-0.094	-0.184
serverType = nginx	-0.029	-0.003	0.004	0.009	-0.062	0.078	-0.458	-0.029	-0.094	1	-0.045
uIrLenght	-0.123	-0.076	0.075	-0.201	-0.165	0.157	0.034	-0.025	-0.184	-0.045	1



MODELLING PHASE – NAIVE BAYES

- Naive Bayes method uses Bayes theorem for probabilistic classification
- Assumption of independence among features given the class label
- Dataset suitability: Variables exhibit low correlations in the correlation matrix

→ Independence assumption does not significantly impact model performance

- RESULTS:

Confusion Matrix

	true FALSE	true TRUE	class precision
pred. FALSE	48408	2406	95.27%
pred. TRUE	0	0	0.00%
class recall	100.00%	0.00%	

Confusion Matrix

	true range1	true range2	true range3	class precision
pred. range1	23694	454	7429	75.04%
pred. range2	0	0	0	0.00%
pred. range3	3271	181	3785	52.30%
class recall	87.87%	0.00%	33.75%	

MODELLING PHASE – LOGISTIC REGRESSION

- Transforms probabilities into a linear connection with predictors
- Coefficients quantify variables' effects on event likelihood
- Logistic regression maximizes likelihood of observed outcomes
- Trained model predicts event probability for new instances
- RESULTS:

Confusion Matrix			
	true FALSE	true TRUE	class precision
pred. FALSE	47995	1858	96.27%
pred. TRUE	411	549	57.19%
class recall	99.15%	22.81%	

Confusion Matrix				
	true range1	true range2	true range3	class precision
pred. range1	26547	536	8613	74.37%
pred. range2	0	0	0	0.00%
pred. range3	418	99	2601	83.42%
class recall	98.45%	0.00%	23.19%	

MODELLING PHASE – DECISION TREE

- Hierarchical structure with nodes representing decisions based on input features
- Splitting criteria: it selects the best features to split the data at each node, creating subsets with similar outcomes
- Predictive power: it can capture complex relationships between features and target variables
- Interpretability advantage
- RESULTS:

Confusion Matrix			
	true FALSE	true TRUE	class precision
pred. FALSE	48400	2009	96.01%
pred. TRUE	2	402	99.50%
class recall	100.00%	16.67%	

Confusion Matrix				
	true range1	true range2	true range3	class precision
pred. range1	26654	549	8333	75.01%
pred. range2	0	0	0	0.00%
pred. range3	311	86	2881	87.89%
class recall	98.85%	0.00%	25.69%	

CONCLUSIONS

- Class Imbalance Challenge:
 - Imbalanced dataset hinders accurate prediction of hidden fraudulent URLs
- Decision Tree Performance:
 - Decision tree model outperformed others
 - URL length plays a significant role in distinguishing genuine and fraudulent URLs
- Differentiating Compromise Types:
 - Successful differentiation of "defacement" and "phishing" compromises
 - Accuracy for the "normal" class remains challenging due to limited representation
- Variable Importance:
 - URL length, "poweredBy," and "serverType" are key factors